MATH 8655: General Topology

Fall 2026

Instructor:	Prof. Zhengchao Wan
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Office:	MSB 308
Office Hours:	Wednesday, 2:00–4:00 p.m., and by appointment
Class Meetings:	Monday, Wednesday, and Friday, 1:00–1:50 p.m. (MSB 111)
Course Website:	https://zhengchao-wan.com/general-topology/

Course Description and Learning Goals

This course is a rigorous introduction to point-set (general) topology. Core themes include topological spaces, continuity, product and quotient constructions, convergence, countability and separation axioms, metrization, compactness, and connectedness. Emphasis is placed on proof writing and the abstract reasoning used throughout analysis, geometry, and algebra.

By the end of the course, students should be able to:

1. work fluently with the principal definitions and constructions of general topology;
2. prove statements by selecting and applying appropriate definitions and theorems;
3. construct examples and counterexamples that distinguish related topological properties;
4. identify hidden assumptions, missing steps, and incorrect conclusions in a proposed proof; and
5. communicate a complete mathematical argument clearly and independently.

Texts and Course Materials

There is no required textbook purchase. The following are recommended references:

- Stephen Willard, *General Topology*.
- James R. Munkres, *Topology*, 2nd edition.

Definitions, notation, and assessment expectations will be established in class and on the course website. All materials needed to prepare for exams will be available without requiring access to a paid AI system.

Assessment and Grading

The course uses two focused midterms rather than one broad midterm. Together with six written homework assignments, this gives students regular opportunities to practice proof writing, receive feedback, and adjust their study strategies before later topics build on the foundations.

Component	Tentative date	Weight
Six LaTeX Homework Assignments	See course schedule	15%
Midterm Exam 1	Friday, September 25	25%
Midterm Exam 2	Monday, November 2	25%
Comprehensive Final Exam	Monday, December 14, 12:30–2:30 p.m.	35%
Total		**100%**

The final exam is scheduled for Monday, December 14, 12:30–2:30 p.m. Midterm dates are tentative until confirmed on the course website and in class.

Exams

Exams are closed-tool assessments: no notes, textbooks, internet access, generative AI, or outside assistance. They assess both mathematical conclusions and the reasoning supporting them.

Problem Bank

One ungraded problem bank, ordered to follow the lecture sequence, will support both midterms and the comprehensive final. Before each exam, the instructor will announce the boundary of the bank eligible for that exam. Every exam problem will be selected from the announced portion.

LaTeX Homework

Six required homework assignments will be given during the semester. Each solution must be typed in LaTeX and submitted for grading and individual feedback. Due dates are listed on the course schedule. Homework is evaluated for mathematical correctness, completeness of reasoning, organization, and clear mathematical writing.

Make-Up Exams

Make-up exams are approved only for serious, documented, unavoidable reasons, including a verified medical emergency, bereavement, or university-sanctioned travel. Notify the instructor in advance whenever reasonably possible and provide timely documentation. Personal travel, work schedules, and convenience are not grounds for a make-up exam.

Grading Scale

Letter grades are assigned according to the following thresholds:

Grade	Criterion
A+	$\geq 97\%$
A	$\geq 90\%$
B+	$\geq 87\%$
B	$\geq 80\%$
C+	$\geq 77\%$
C	$\geq 70\%$
D+	$\geq 67\%$
D	$\geq 60\%$
F	$< 60\%$

Generative AI Policy

Category 2: Limited AI Use Permitted. AI tools may be used as study resources and for homework. AI tools are not permitted during exams. Students are responsible for checking AI-generated content, which can be inaccurate, offensive, or biased. Because AI will not be available during exams, students should be prepared to solve problems independently without AI assistance. Over-reliance on AI tools may diminish your ability to reach the required level of mastery.

Attendance, Participation, and Communication

Regular attendance and active participation are expected, although attendance is not separately graded. Students are responsible for announcements and material covered in class. If an absence affects an exam, contact the instructor as early as possible. Questions and concerns are welcome by email, in office hours, or by appointment.

Students may not make audio or video recordings of course activity without permission, except when recording is part of an approved accommodation.

University Policies and Support

Academic integrity is fundamental to university work. Any effort to gain an advantage not available to all students is dishonest, whether or not the effort is successful. When in doubt about collaboration, attribution, AI use, or any other form of assistance, ask the instructor before submitting graded work.

Mizzou's current statements on academic integrity, accommodations, religious observance, intellectual pluralism, student privacy, nondiscrimination, and student well-being are maintained on the Office of the Provost's syllabus-information page and in Canvas under *Supports & Policies*. These university policies apply to this course.

Students seeking disability-related accommodations should contact the Disability Center to establish an Accommodation Plan and notify the instructor as soon as possible. Students seeking mental-health support may contact the Mizzou Counseling Center and well-being resources; phone support is available at 573-882-6601.

Course policies and dates may be adjusted when necessary. Material changes will be announced in class and posted on the course website.